# Applied SVAR Identification

## Day 3: Identification II — Long-Run Restrictions

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## Course Roadmap

| Day | Topic |
|---|---|
| 1 | VAR Foundations: Motivation, Estimation, Specification, Identification & Tools |
| 2 | Identification I: Short-Run Restrictions |
| 3 | Identification II: Long-run Restrictions |
| 4 | Identification III: Sign, Magnitude, and Narrative Restrictions |
| 5 | Identification IV: Instruments |

# Learning Objectives

By the end of today, you should be able to:

1. Explain what long-run identification is and how it solves the identification problem.
2. Understand where the long-run multiplier matrix and the Blanchard–Quah identification strategy comes from.
3. Implement long-run identification in MATLAB using the ACB Toolbox.
4. Apply long-run restrictions to replicate Blanchard–Quah and Bjørnland (2009).

Part 1

# Long-Run Zero Restrictions

Restricting the long-run impact of structural shocks

# When Short-Run Restrictions Are Problematic

In some applications, *no* strict contemporaneous ordering is economically defensible.

**Example: Interest Rate and Exchange Rate**:

- If interest rate is ordered *before* exchange rate then the exchange rate cannot respond to a monetary policy shock within the quarter — unrealistic for a floating exchange rate.
- If exchange rate ordered *before* interest rate then the central bank cannot react to exchange rate movements within the month/quarter — also unrealistic.

**New Idea**: Since monetary policy only impacts exchange rates in the short-run, restrict the *long-run cumulative effects* of structural shocks, instead of the short-run contemporaneous effects.

**Key Idea**

Long-run zero restrictions are motivated by economic theory about which shocks can have 'permanent' effects — rather than which variables react first.

## Identification Critiques

- Cooley and LeRoy (1985) defended the Cowles Commission approach as trying to incorporate Keynesian macroeconomics into macroeconometric models, and then criticized the early use of VARs as being 'atheoretical macroeconometrics' despite the claims of Sims and others.
- The key issue was that *ad hoc* recursive ordering was commonly unjustifiable.
- Bernanke (1986), Blanchard and Watson (1986) and others began to apply non-recursive orders that still satisfied the order condition.
- Blanchard & Quah (1989) idea: Instead of applying restrictions contemporaneously, can we incorporate long-run economic theory into SVARs?

## The Long-Run Effect of a Shock

**Recall**: Using results from Day 1, the MA representation for a VAR(1) (excluding constant) is:

$$y_t = \sum_{j=0}^{\infty} A_1^j \, e_{t-j}$$

Since $e_t = B_0^{-1}\varepsilon_t$, if the VAR is stable, then the cumulative long-run effect of structural shocks, $\varepsilon_t$, on $y_t$ is:

$$C = (I - A_1)^{-1} \, B_0^{-1}$$

**Interpretation**: Element $[C]_{ij}$ gives the total accumulated long-run effect of shock $j$ on variable $i$.

**Identification Insight**: Imposing $[C]_{ij} = 0$ means that structural shock $j$ has no cumulative long-run effect on variable $i$ – can also use any other constant.

# The Blanchard–Quah Restriction

## Blanchard-Quah Long-Run Restrictions

**Blanchard & Quah (1989)** identify the SVAR by making the long-run impact of structural shocks, $C$, lower triangular.

Note: This gives exactly $\frac{n(n-1)}{2}$ restrictions $\rightarrow$ satisfies the order condition!

Example: Assume $y_t = (\Delta GDP, u_t)'$ are driven by an aggregate supply shock, $\varepsilon_t^{Supply}$, and an aggregate demand shock, $\varepsilon_t^{Demand}$. The long-run cumulative impact is:

$$\underbrace{\begin{bmatrix} \Delta GDP_{t,t+\infty} \\ u_{t,t+\infty} \end{bmatrix}}_{y_t} = \underbrace{\begin{bmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \end{bmatrix}}_{C} \underbrace{\begin{bmatrix} \varepsilon_t^{Supply} \\ \varepsilon_t^{Demand} \end{bmatrix}}_{\varepsilon_t}$$

We can impose that the long-run level of output is entirely driven by aggregate supply shocks by setting $c_{12} = 0$.

## The BQ Restriction in Practice

In practice, we implement BQ restrictions in three steps:

1. The *long-run covariance matrix*, $\Omega$, is given by

$$\Omega = CC' = (I - A_1)^{-1} \underbrace{(B_0^{-1})(B_0^{-1})'}_{\hat{\Sigma}} ((I - A_1)^{-1})'$$

2. Take the Cholesky decomposition of $\Omega$ so that:

$$\Omega = PP'$$

   where $P$ is the lower-triangular Cholesky factor of $\Omega$.

3. Use this to recover the impact matrix:

$$C = (I - A_1)^{-1} B_0^{-1} = P \quad \Rightarrow \quad \boxed{\hat{B}_0^{-1} = (I - A_1) P}$$

Done in ACB Toolbox using ‘VARopt.ident = 'long';‘

## Short-Run vs Long-Run: A Comparison

| | **Short-Run (Cholesky)** | **Long-Run (BQ)** |
|---|---|---|
| Key paper | Sims (1980) | Blanchard & Quah (1989) |
| Restriction | $B_0^{-1}$ lower triangular | $B_0^{-1}$ recovered after making $C$ lower triangular |
| Economic content | Contemporaneous causal order | Long-run neutrality |
| ACB Toolbox | ‘VARopt.ident = 'short';‘ | ‘VARopt.ident = 'long';‘ |

Both strategies achieve **exact identification** with $\frac{n(n-1)}{2}$ restrictions. The difference is *where* those restrictions are placed: on the contemporaneous impact, or on the long-run cumulative effect.

# Long-Run Restrictions: Pros and Cons

## Pros

- Grounded in long-run theory which is often less controversial among economists
- Allows contemporaneous interactions that short-run restrictions forbid
- Can resolve ordering sensitivity in key applications

## Cons

- Requires accurate estimation of $C$: $C$ may be poorly estimated in short samples – potentially problematic with current macroeconomic data
- Can become unstable when series are extremely persistent (especially in short-samples)

Part 2

# Blanchard & Quah (1989)

“What are the Dynamic Effects of Aggregate Demand and Supply Disturbances?”

# Blanchard & Quah (1989, AER): Overview

**Question**: What are the dynamic effects of aggregate demand and supply disturbances on output and unemployment?

**Motivation**: Standard VARs cannot distinguish demand from supply shocks — both shift output and unemployment contemporaneously. Yet their *long-run* effects on output differ fundamentally: supply shocks are permanent, demand shocks are transitory.

**Contribution**: Separate permanent (supply) from transitory (demand) output disturbances using a single long-run neutrality restriction, without imposing any contemporaneous ordering.

**Identification**: The long-run output level is driven by supply-side (total factor productivity (TFP)) shocks. Demand shocks have no permanent effect on the level of output. In a 2-variable system, this yields one long-run identifying restriction:

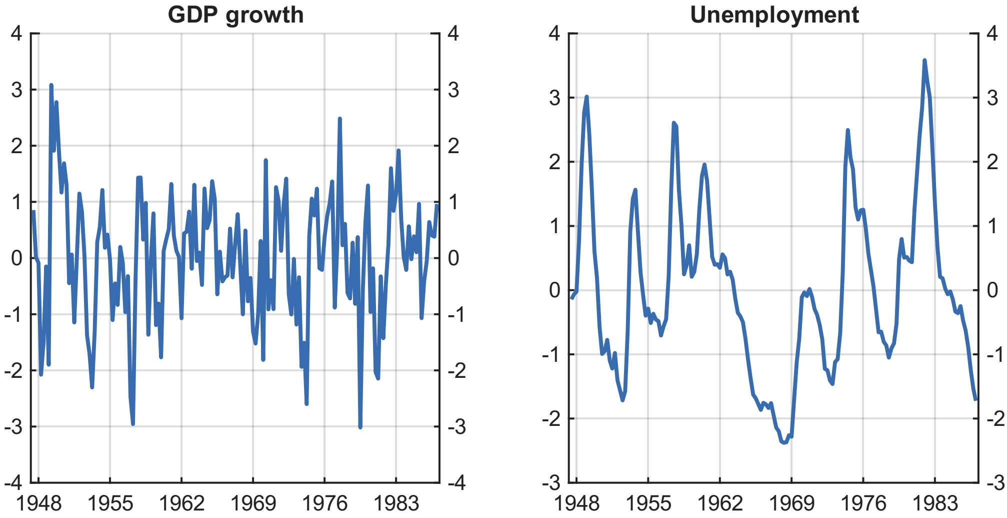
$$[C]_{1,2} = 0$$

# Blanchard & Quah (1989): The VAR

- **Quarterly VAR**, 1950:Q2–1987:Q4, $p = 8$ lags.
- **Two variables** in long-run order:
  1. $\Delta\text{GDP}_t$ = Real GDP growth (first difference of log)
  2. $u_t$ = Unemployment rate (level)

  Note: GDP enters in *growth rates* to make the model stable; the long-run restriction zeros out the cumulative GDP-growth response to the demand shock.
- **Two structural shocks**:
  1. **Demand shock**: *no* long-run effect on output (e.g., fiscal stimulus, monetary policy)
  2. **Supply shock**: permanent effect on output (e.g., TFP)

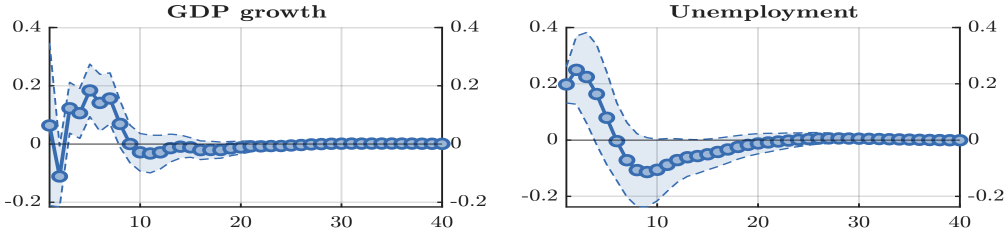
# Blanchard & Quah (1989): Data



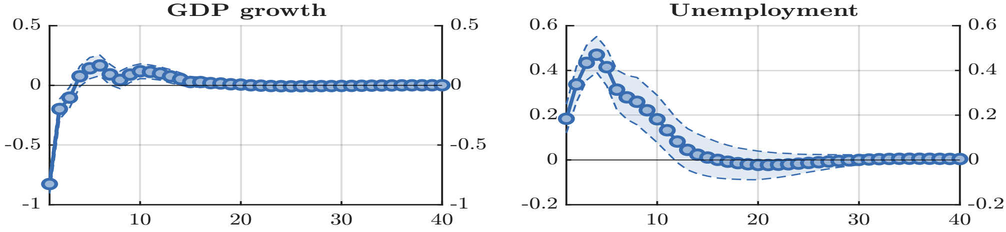

Ex: Identify the main business cycle episodes visible in the data

# Blanchard & Quah (1989): Impulse Responses to Each Structural Shock

## (a) Supply shock

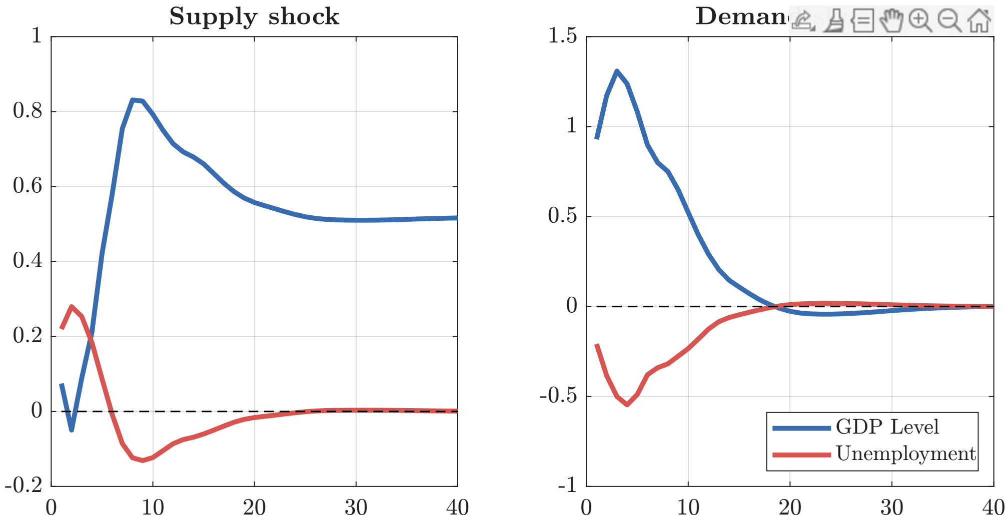



## (b) Demand shock




Ex: Interpret the results.

# Blanchard & Quah (1989): Cumulative IRFs




Ex: Is the long-run restriction satisfied?

# Key Takeaways: Blanchard & Quah (1989)

1. **Long-run neutrality identifies shocks**: A single theoretical restriction, i.e., demand is neutral in the long run, is enough to exactly identify two structural shocks in a 2-variable system.
2. **Supply shocks drive long-run output**: Supply disturbances permanently raise output and initially raise unemployment (consistent with short-run labour market friction).
3. **Demand shocks drive the business cycle**: Demand disturbances produce the typical boom-bust cycle — output rises then reverts; unemployment falls then recovers. No permanent effect on output.
4. **Be careful when plotting IRFs!**: The restriction makes the demand shock's cumulative GDP IRF converge to zero...

Part 3

# Bjørnland (2009)

"What is the impact of monetary policy on the exchange rate?"

# Bjørnland (2009, JIE): Overview

**Question**: Does a monetary policy tightening cause the real exchange rate to appreciate or depreciate on impact?

**Motivation**: Dornbusch's (1976) overshooting hypothesis asserts that an increase in the interest rate should cause the nominal exchange rate to appreciate instantaneously, and then depreciate in line with uncovered interest parity (UIP). However, recursively identified SVARs show that following a monetary tightening:

1. Exchange rate depreciates — called **exchange rate puzzle**; or
2. Exchange rate appreciates after a lag — called **delayed overshooting**

**Contribution**: Show that the puzzle is an artifact of incorrect identification! Resolve the exchange rate puzzle by using a long-run neutrality restriction, thereby allowing simultaneous feedback between the interest rate and the exchange rate.

**Identification**: Monetary policy has no permanent effect on the real exchange rate. Implemented via long-run zero restriction.

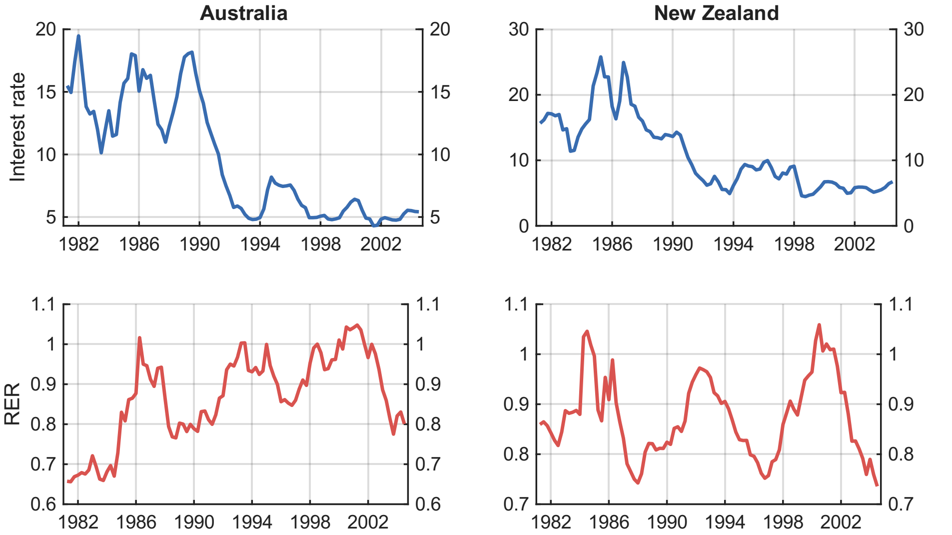
# Bjørnland (2009): The VAR

- **Quarterly SVAR**, 1982:Q1–2004:Q4; Australia and New Zealand (she has more in the paper). $p = 3$ lags, constant + linear trend.

- **Five variables** in long-run order:
  1. $i_t^*$ = Foreign interest rate (level)
  2. $\text{gdp}_t$ = Real GDP (log)
  3. $\pi_t$ = CPI inflation (level)
  4. $i_t$ = Domestic interest rate (level)
  5. $\Delta q_t$ = Real exchange rate (log difference) – domestic/foreign $\rightarrow$ decrease is an appreciation and vice–versa.

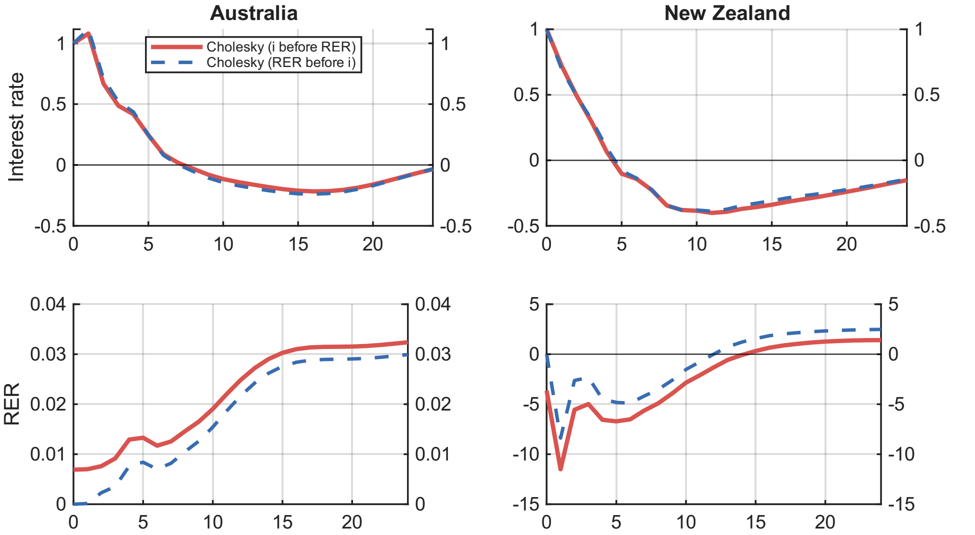
- **One structural shock of interest**: monetary policy shock (shock 4) — identified as having no permanent effect on the level of the real exchange rate.
  Note: In the paper she uses short-run zero restrictions of $i_t^*, \text{gdp}_t$ and $\pi_t$ to a monetary policy shock, but here we only use long-run zero restrictions as in BQ.

# Bjørnland (2009): Data, 1982–2004



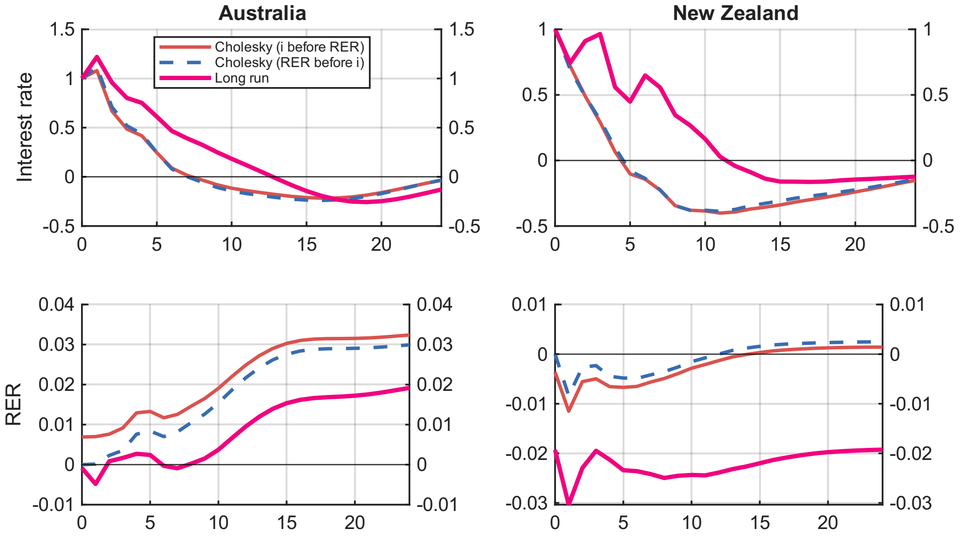

Ex: How do interest rates and real exchange rates co-move across Australia and NZ?

# Bjørnland (2009): Cholesky IRFs




Ex: Does the exchange rate appreciate or depreciate on impact? Is this consistent with Dornbusch overshooting?

# Bjørnland (2009): Identification Matters




Ex: Are the long-run identified responses consistent with Dornbusch overshooting?

## Key Takeaways: Bjørnland (2009)

1. **Dornbusch was right**: Under long-run identification, a monetary policy tightening produces an *on-impact appreciation* of the real exchange rate in all four countries, consistent with Dornbusch's overshooting hypothesis.
2. **Identification drives results**: The exchange rate puzzle arises from inappropriate Cholesky restrictions — not from the data. Changing identification changes the conclusion.
3. **Long-run neutrality is the key assumption**: Monetary policy cannot permanently alter the real exchange rate. This is the economically grounded restriction that resolves the puzzle.
4. **Results are robust**: In her paper, the long-run identification gives similar results across all countries. Our results are a bit different because she combines short-run and long-run restrictions which is not supported by the ACB Toolbox (see her paper for details).

# Main Takeaways

1. **Long-Run Restrictions**: Place restrictions on the cumulative long-run impact of a shock. Still need to satisfy the order condition, and should be in line with economic theory.
2. **Blanchard-Quah Approach**: Make the cumulative long-run effects lower triangular. In their two-variable model, this satisfies the order condition, and is supported by macroeconomic theory.
3. **Practical point**: BQ is fine in theory, but requires a 'long-run' to exist in the data. This may be problematic in short samples.

# Preview of Day 4

1. **Sign restrictions** — move away from exact identification and use theory to provide a set of SVARs that satisfy qualitative sign restrictions (Uhlig, 2005).

2. **Magnitude restrictions** — refine the set of feasible SVARs using plausible magnitudes of certain parameters, e.g., elasticities (Kilian and Murphy, 2012).

3. **Narrative restrictions** — refine the set of feasible SVARs using known events, e.g., October 1979, the month of the Volcker disinflation announcement (Antolin-Diaz and Rubio-Ramirez, 2018).

Thank you!